Daiwei Chen

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Research Focus: LLM Factuality, Safety & Evaluation

Ph.D. researcher studying LLM factuality, robustness, and the safety–capability trade-off. Research spans RAG evaluation, pluralistic alignment, and LinkedIn Responsible AI, with an emphasis on both reliability and usefulness.

Education

University of Wisconsin-Madison — Madison, WI
Ph.D. in Electrical and Computer Engineering — *Sep 2023 – Present*
Advisor: Ramya Korlakai Vinayak

University of Pennsylvania — Philadelphia, PA
M.S. in Systems Engineering — *Aug 2021 – May 2023*

Zhejiang University — Hangzhou, China
B.S. in Psychology — *Sep 2017 – Jun 2021*

Industry Research Experience

LinkedIn — Mountain View, CA

Responsible AI - Research Intern — Jun 2026 – Sep 2026

- Developed **Multi-Teacher On-Policy Distillation (MOPD)** to address the **alignment tax**: the trade-off between safety alignment and model capabilities in LLM post-training.

Generative AI - Research Intern — May 2025 – Aug 2025

- Developed **GTI** for generative candidate retrieval; improved offline **Precision@5 by 21.6% relative** to vanilla SFT under the Good Match criterion on an industrial dataset.

Selected Research & Publications

Is Conformal Factuality for RAG-based LLMs Robust? Novel Metrics and Systematic Insights
Y. Chen, D. Chen, S. M. Chikodikar, C. H. Yin, R. K. Vinayak
Under review | *Preprint, arXiv:2603.16817, 2026*
Utility-aware metrics reveal when factuality filtering sacrifices useful information or fails under shift.

One Model, Many Modes: Evolving System Prompt Ensembles for Creative Generation
D. Chen, A. He, B. Yao, J. Hu, R. K. Vinayak
Under review
Prompt ensembles improve novelty and diversity while preserving quality across four tasks and three models.

PAL: Sample-Efficient Personalized Reward Modeling for Pluralistic Alignment
D. Chen, Y. Chen, A. Rege, Z. Wang, R. K. Vinayak
ICLR 2025 | *NeurIPS 2024 Workshop (**Spotlight**)* | *ICML 2024 Workshop (**Oral**)*
Personalized reward models represent diverse human preferences and adapt to new users with few examples.

Grounded Token Initialization for New Vocabulary in LMs for Generative Recommendation
D. Chen, Z. Fu, C. Jiang, H. Zhang, R. Zhou, T. Wang, C. Yao, G. Li, R. Cai, Y. Cao, R. Jiang, F. Borisyuk, J. Shen, J. Wu, R. K. Vinayak
EMNLP 2026 *Main Conference*
Spectral diagnostics reveal token-embedding collapse; grounded initialization improves generative retrieval.

Compositional Self-Improvement
C. Shin, D. Chen, J. Cooper, B. Lake, F. Sala, R. K. Vinayak
ICML 2026 *– 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents*
Composing reliable subproblem predictions enables self-training beyond the original supervision regime.

Service & Awards

Reviewer: NeurIPS 2026, ICLR 2026, CVPR 2026, EACL 2026, COLM 2026.
Awards: Lambda.ai Research Grant Sponsorship (2025); UPenn Outstanding Research Award (2023).